Problem 5

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Problem 1 *Evaluate each of the following limits. Possible answers include a number, $+\infty$, $-\infty$ and "Does Not Exist" (DNE). Make sure to state the form of the limit. Justify your answer.*

(a) $\lim\limits_{x\to 3^-} \dfrac{x^2-3}{x^2-x-6}$

(b) $\lim\limits_{x\to 5^+} \dfrac{x^2+6}{x^2-3x-10}$

(c) $\lim\limits_{x\to 1} \dfrac{4-x}{x^2-2x+1}$

(d) $\lim\limits_{x\to 2} \dfrac{-e^x}{(2-x)^3}$

(e) $\lim\limits_{x\to 1} \dfrac{\sin x}{\sqrt{2-x^2}-1}$